

Model Paper - Database Management Systems

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Q1 (20 marks)

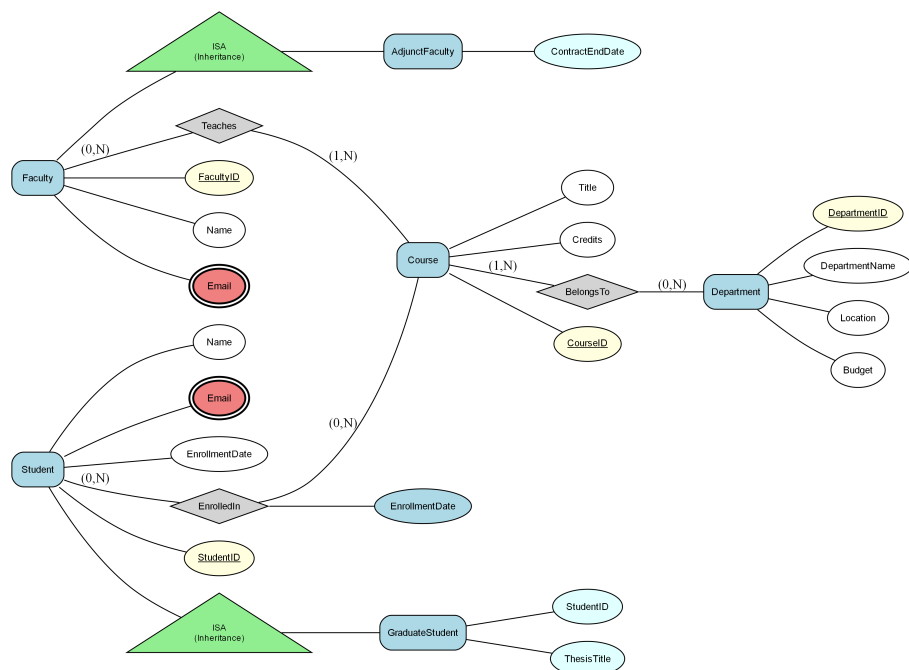
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university database system is designed to manage information about faculty members, courses, students, and departments. The Faculty entity has attributes: FacultyID, Name, and Email. The Course entity includes CourseID, Title, and Credits. The Student entity has attributes such as StudentID, Name, Email, contains StudentID, Name, and EnrollmentDate. The Department entity has attributes such as DepartmentID, DepartmentName, Location, consists of DepartmentID, Name, and Budget. Faculty members may teach multiple courses, and courses can have many students enrolled. Each course belongs to one department. There is a specialization where some faculty can be 'Adjunct Faculty' with an additional attribute: ContractEndDate, and some students can be 'Graduate Students' with an additional attribute: ThesisTitle. A Faculty can be related to many Courses through the 'Teaches' relationship (Faculty participation is optional). A Student can be related to many Courses, and a Course can be related to many Students through the 'EnrolledIn' relationship (Student participation is optional, Course participation is optional). Many Courses can be related to one Department through the 'BelongsTo' relationship (Department participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E, AE \rightarrow C\}$.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is implementing a new database system to manage its courses, students, and faculty information. The schema includes tables for Courses(course_id: string, course_name: string, credits: integer), Students(student_id: string, name: string, major: string), and Faculty(faculty_id: string, name: string, department: string). The university requires that students can enroll in multiple courses and faculty can teach multiple courses. As part of this system, proper integrity constraints must be applied to maintain data accuracy and consistency.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment:

```
``java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery(); ``
```

 Which type of JDBC statement is used in the code segment shown above? Briefly

explain when this type of statement will be used. (2 marks)

c) Briefly explain how primary and foreign keys can be used to maintain referential integrity in this university database scenario. (2 marks)

d) Briefly explain the importance of implementing data types and constraints in the database design for the university's schema. (3 marks)

e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)

iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Library: Book (bookId: int, title: varchar(100), author: varchar(50), publicationYear: int) Member (memberId: int, name: varchar(100), email: varchar(50), phone: varchar(15)) Loan (loanId: int, bookId: int, memberId: int, loanDate: date, returnDate: date) Fine (fineId: int, memberId: int, amount: real, paymentStatus: varchar(20)). The 'Book' table stores information about books available in the library, including unique book ID, title, author, and publication year. The 'Member' table keeps details of library members, containing their ID, name, email, and phone number. The 'Loan' table records transactions between members and books, with information on loan IDs, book IDs, member IDs, loan dates, and return dates. The 'Fine' table tracks fines incurred by members, providing details such as fine IDs, member IDs, fine amounts, and payment statuses. The 'Member' table holds details about library members, such as their unique member ID, first name, last name, email, phone number, and address. The 'Loan' table manages book loans, with each loan having a unique ID and being associated with a specific book and member. It records the loan date, due date, and return date. The 'Fine' table contains information about fines incurred by members for late returns or other penalties. It includes a fine ID, member ID, fine amount, and payment status.

a) Write SQL Queries to perform the following: i. Find the member's name and email who currently has a book on loan.

i. Find the member's name and email who currently has a book on loan. (4 marks)

ii. Find the member who has the highest total amount compared to all other members. Find the memberId and name. (6 marks)

iii. Find the title and name from Book, Member, and Loan where Loan.loanId is not null. (7 marks)

- b) b) Create a function to calculate the total fine amount for a given member. This function should account for member with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)
- c) c) Create a trigger that automatically updates the 'amount' column in the 'Fine' table whenever a new fine record is added, or an existing fine record is updated. The 'amount' column should contain the total fine amount for each member. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by member. The member pays only 50% of the total fine amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)